

ARENA CONNECTIVITY

Bluetooth broadcast, mesh relay and offline venue network research

Technical assessment, capacity model, cost plan and DIY build guide



RECOMMENDATION

**Build a fixed powered Loc8 relay layer for the venue.
Use BLE for discovery and broadcast, then use a hybrid
peer-to-peer transport for reliable two-way phone data.**



DOCUMENT MAP

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HOW TO USE THIS REPORT

Sections 1 to 5 support product and investment decisions. Sections 6 to 11 support engineering, purchasing and venue deployment. The appendices provide a direct answer script, glossary, assumptions and source register.

Document status

Item	Value
Company	Loc8
Purpose	Research for offline friend-finding and venue connectivity
Version	1.0
Research date	21 July 2026
Geographic focus	United Kingdom, with platform notes for iOS and Android
Classification	Confidential working research

This is an engineering research document, not legal advice, a spectrum licence, a venue safety plan or a conformity assessment. Product launch requires review by qualified RF, compliance, electrical and venue professionals.

WHAT LOC8 SHOULD BUILD

1. Executive decision

Loc8 can cover a large arena, but not by making one Bluetooth transmitter exceptionally powerful. The dependable design is a distributed network of fixed, powered nodes placed around the venue. Each node covers a local zone. Nodes exchange small control messages through Bluetooth Mesh or a wired/IP backhaul, while phones receive broadcasts and use the best available peer-to-peer transport for two-way user data.

INDOOR ARENA NODES 6-8 Initial planning range	STADIUM NODES 8-16+ Include stands and concourses	MESH ELEMENTS 32,767 Protocol address ceiling, not crowd capacity	PASSIVE RECEIVERS 80,000+ Broadcast audience is not connection-limited
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RECOMMENDED PRODUCT POSITION

Sell Loc8 as an offline-capable friend-finding system with optional venue infrastructure. Do not describe it as an emergency communications system or promise a fixed 100 m range inside crowds.

The decision in one table

Question	Answer	Meaning for Loc8
Can one device cover an arena?	Not reliably.	Use multiple low-power cells with overlap.
Can thousands of phones hear one beacon?	Yes, in principle.	Advertising is one-to-many and does not create a connection per listener.
Can thousands maintain two-way BLE links to one node?	No.	Direct link counts are measured in tens, with less usable in production.
Can mesh relay over a large area?	Yes.	Fixed relay nodes repeat short messages over multiple hops.
Should attendees' phones be the mesh backbone?	No.	Background limits, batteries and unpredictable movement make it unreliable.
What should carry richer two-way data?	A hybrid transport.	BLE discovery plus Wi-Fi Aware, Nearby Connections or venue Wi-Fi where supported.

Recommended Loc8 hardware family

Unit	Role	Prototype form
Loc8 Anchor	Zone ID and passive proximity	Commercial beacon or nRF52840
Loc8 Relay	Powered BLE broadcast and mesh relay	nRF21540 Development Bundle in IP-rated enclosure
Loc8 Gateway	Local control, monitoring and venue interface	Relay plus Ethernet/PoE host
Loc8 Portable	Temporary battery-powered event coverage	Relay with protected battery pack and timed deployment

CAPACITY STARTS WITH THE OPERATING MODE

2. Four technologies that must not be confused

The phrase 'Bluetooth device' can mean several very different things. The number of people served changes radically depending on whether a device advertises, connects, relays mesh messages or asks phones to relay.

Mode	What happens	Capacity concept	Best Loc8 use
BLE broadcast	A node transmits short advertisements; phones listen.	No per-listener link. Effectively unlimited passive listeners, constrained by radio reception and phone scanning.	Zone identity, beacons, short alerts and discovery.
Direct BLE connection	Central and peripheral maintain timed two-way link state.	nRF52 controllers support about 20 in the cited production stack; practical count is lower when traffic, latency and reliability matter.	Configuration, maintenance and small-group accessories.
Bluetooth Mesh	Addressed elements publish messages; relay nodes retransmit selected traffic.	32,767 element protocol ceiling. Real capacity depends on traffic rate, density, relays and RF conditions.	Fixed venue control plane and small state updates.
Phone relay / opportunistic mesh	Phones scan, receive and re-advertise or forward.	No trustworthy crowd figure. Mobile OS scheduling, battery and movement make availability probabilistic.	Optional enhancement only, never the sole venue backbone.

THE HONEST ANSWER TO 'HOW MANY PEOPLE?'

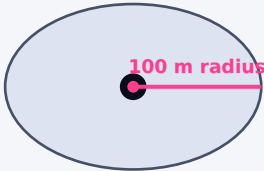
A broadcast beacon can be heard by every compatible phone in range without making a separate connection. A direct two-way node can handle only a small number of simultaneous connections. A formal Bluetooth Mesh can address 32,767 elements, but that number is not the same as 32,767 attendees exchanging live location data.

Why an arena audience is not one network

A crowd of 20,000 phones is best treated as a very large set of receivers and peer endpoints, not as 20,000 infrastructure routers. The infrastructure should carry sparse, prioritised control data. Private location updates should remain local to crews or use a higher-capacity peer-to-peer or IP path.

WHAT 100 METRES REALLY MEANS

3. Range and arena coverage



A diagram showing a light blue circle with a black center. A red line segment extends from the center to the right edge of the circle, labeled "100 m radius".

What 100 metres means

200 m	diameter
31,416 m²	circle area
7.76 acres	same area
~1 football pitch	end-to-end distance

Geometry is not RF performance: bodies, walls, steel, antenna height and interference shrink usable range.

A 100 m claim normally describes an unobstructed line-of-sight test under stated power, antenna and receiver conditions. It is not a promise that a phone in a pocket will receive the signal through a dense crowd at 100 m.

Range drivers

Driver	Improves range when...	Hurts range when...
Transmit power	Within legal EIRP and hardware limits	Power is reduced, detuned or blocked
Receiver sensitivity	Coded PHY or a strong scanner is supported	Ordinary phones use less sensitive modes
Antenna	Clear, correctly tuned and elevated	Near metal, water, bodies or inside a metal box
Path	Line of sight with good height	Concrete, steel seating, stages and people obstruct it
Interference	Channels have low utilisation	Wi-Fi, Bluetooth and production radios congest 2.4 GHz
Packet repetition	Several advertising opportunities exist	Duty cycle is too low or phones scan infrequently

Planning ranges for Loc8

Environment	Initial cell radius	Design implication
Open field, node elevated	60-100 m	Validate beyond 100 m, but do not base safety-critical promises on it.
Arena bowl with sightlines	35-70 m	Use overlapping sectors around the bowl.
Dense floor crowd	15-40 m	Elevate nodes and add floor/perimeter coverage.
Concrete concourse	10-30 m	Treat bends and fire doors as separate zones.
Backstage / service areas	10-25 m	Use dedicated nodes; do not expect the bowl network to penetrate.

These are conservative planning allowances, not measured Loc8 results. A site survey and pilot must replace them before any contractual coverage commitment.

BUY, PROTOTYPE OR BUILD

4. Devices that exist now

Three product categories can support Loc8 research. They solve different problems and should not be compared only by advertised distance.

Device / platform	Published capability	Indicative price	Loc8 judgement
Minew MBM01 beacon	Vendor claims long-range variants, including 210 m and up to 600 m under stated configurations.	Request quote	Useful for fast beacon range experiments. Not a complete two-way arena network.
Blue Charm BC04P	Vendor claims about 100 m to legacy smartphones and 300 m with LE Coded PHY.	Low-cost beacon class	Useful for inexpensive trials. Phone support and conditions determine achieved range.
Listen Technologies Auri TX2N	Two-radio Auracast transmitter, PoE and repeater mode; vendor describes over 100 m coverage.	About £1,983 ex VAT for Dante listing	Professional broadcast audio, not a general Loc8 data mesh. Relevant as proof that managed BLE broadcast can cover venues.
Nordic nRF52840 DK	Multiprotocol BLE SoC development platform.	About £36.26 ex VAT	Good low-cost firmware development board; lower link budget than FEM bundle.
Nordic nRF21540 DB	nRF52840 plus front-end module, two onboard antennas and separate evaluation board.	£63.64 ex VAT, about £76.37 inc VAT	Best recommended long-range DIY prototype platform.

What the nRF21540 adds

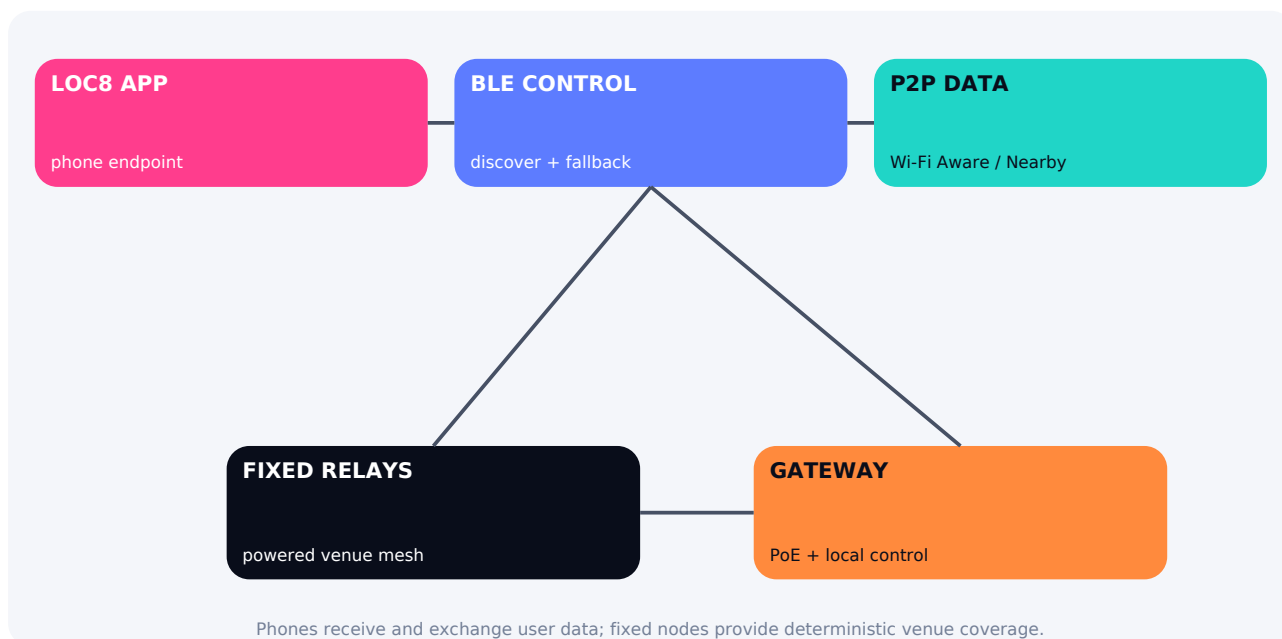
Nordic specifies up to +20 dBm transmit output and about 13 dB receive gain for the nRF21540 front-end module, with tunable output up to +21 dBm in some configurations. Nordic describes a theoretical 6.3 to 10 times range extension relative to an unassisted radio. 'Theoretical' matters: legal EIRP, antenna gain, propagation and receiver capability still govern the real result.

PROCUREMENT CAUTION

The Development Bundle includes a full nRF21540 DK and a separate nRF21540 evaluation kit. The simplest prototype uses the complete DK. The evaluation kit requires compatible host hardware and a special cable that Nordic says is not supplied.

A RELIABLE FIXED LAYER PLUS A FLEXIBLE PHONE LAYER

5. Recommended Loc8 architecture



Layer 1: phone experience

The Loc8 app owns crew identity, privacy rules, maps, friend radar, lost mode and user consent. Each phone is primarily an endpoint. It can scan for signed Loc8 zone advertisements and exchange crew data through a supported peer-to-peer transport.

Layer 2: BLE discovery and control

Short advertisements identify zones, provide a time reference, announce available services and carry tiny signed status flags. BLE remains the broad compatibility layer. It should not carry continuous high-frequency GPS positions for an entire crowd.

Layer 3: higher-capacity peer-to-peer data

On supported Apple devices, Wi-Fi Aware offers direct discovery and secure peer-to-peer links. Apple's current technical note states that iOS 26 introduced Wi-Fi Aware on iPhone 12 and later. Android provides Wi-Fi Aware from Android 8.0 where the device hardware supports it. Google Nearby Connections is another offline abstraction that uses Bluetooth, BLE and Wi-Fi and encrypts connections. Platform support must be detected at runtime, with BLE and venue Wi-Fi fallbacks.

Layer 4: fixed venue relays

Powered nodes form a sparse Bluetooth Mesh or use Ethernet/PoE to the gateway. They repeat small control messages, publish health status and provide known RF positions. Where cabling exists, use it for backhaul because wired links reduce RF traffic and simplify monitoring.

Layer 5: local gateway

A gateway runs without internet, manages zones and keys, records node health, accepts authorised venue commands and exposes a local API to Loc8 services. Internet connectivity can synchronise configuration, but the core event operation should survive its loss.

RANGE BY HOPS, NOT AMPLIFICATION

6. How Bluetooth Mesh relaying works

Mesh extends reach by repeating messages over hops



It does not amplify one signal. TTL limits hop count; relay settings and traffic rate control congestion.

A source publishes a mesh message. Nearby relay nodes that receive it may retransmit it. Another relay receives the retransmission and may forward it again. A Time To Live value is reduced on each hop, preventing messages from circulating indefinitely. Standard Bluetooth Mesh uses managed flooding; newer Directed Forwarding features can establish more efficient paths for suitable deployments.

The important mesh numbers

Number	What it actually means	What it does not mean
32,767 elements	Maximum unicast address space for normal mesh elements.	A guarantee that 32,767 busy phones will work in one arena.
6,000 nodes	Scale reported by Bluetooth SIG for real commercial deployments.	A universal performance specification for any message rate or venue.
20 connections	Approximate absolute controller limit cited for nRF52840 production Bluetooth stacks.	Twenty thousand passive broadcast listeners.
Unlimited listeners	Broadcast does not allocate a connection for each receiver.	Every phone will hear every packet in RF shadow or while the OS is not scanning.

Relay traffic discipline

Control	Recommended starting point	Reason
TTL	3 to 5	Enough for a compact venue without excessive retransmission.
Relay retransmits	Low, then tune from measurements	More repeats improve probability but multiply congestion.
Publish interval	Seconds, not continuous streaming	Mesh is efficient for sparse state, not crowd-scale telemetry.
Payload	Identifiers and deltas	Short packets consume less airtime.
Group addressing	Per stand, concourse, floor and service area	Avoids sending every message to every node.
Backhaul	Ethernet/PoE where possible	Moves management traffic off 2.4 GHz.

CAPACITY MODEL FOR LOC8

7. How many people can the system support?

SHORT ANSWER

The mesh element ceiling is 32,767, but that is the wrong crowd metric. A Loc8 venue network can broadcast to an audience of 80,000 or more because receiving does not create connections. The practical two-way capacity depends on message size, frequency, peer grouping, phone behaviour and the non-BLE data path.

Capacity by use case

Use case	Realistic interpretation	Planning rule
Zone beacon received by attendees	Potentially every compatible phone in range.	Design RF overlap and packet repetition; no sign-in connection required.
Crew of 2-20 sharing positions	Plausible peer group, subject to phone OS and transport.	Rate-limit updates and transmit movement deltas.
Node maintenance connections	A handful of authorised engineering devices.	Keep below controller maximum with operational margin.
Arena-wide lost-person alert	Tiny signed message to zones, then phones display locally.	Use group publication and repeat with expiry.
Every attendee sending GPS every second	Not appropriate for Bluetooth Mesh.	Use local crew exchange, venue Wi-Fi/cellular, or much lower rates.

Illustrative traffic calculation

Assume a 12-person crew, each sending a compact 40-byte encrypted position update every 5 seconds. The application payload alone is 96 bytes per second for the crew before protocol overhead and retransmissions. That is manageable. If 20,000 attendees each sent the same update arena-wide, the application payload would be 160,000 bytes per second before overhead, collisions and repeats, and the topology would be unsuitable. The solution is scope: crew-local exchange, zone filtering and event-driven updates.

Policy	Example
Movement threshold	Send only after moving 5-10 m or crossing a zone.
Adaptive interval	5-10 seconds when moving; 30-60 seconds when stationary.
Crew scope	Do not flood private positions arena-wide.
Expiry	Discard old positions after a defined freshness window.
Priority	Lost mode and safety information outrank routine location updates.

MOBILE OPERATING SYSTEMS CONTROL THE RADIO SCHEDULE

8. Why phones should not be the primary relays

Phone relaying is technically possible while an app is active, but the app does not control the radio continuously. Apple documents that background scanning can slow, duplicate discoveries can be coalesced, advertising frequency can decrease and an app can be suspended or terminated. Android offers background mechanisms, but permissions, device variations, power management and OEM policies still affect behaviour.

Risk	Failure mode in a live crowd	Mitigation
Background scheduling	Relay misses or delays packets when the app is not foregrounded.	Use fixed relays; phone relay only as opportunistic bonus.
Battery pressure	Users disable Bluetooth or the app after high drain.	Duty-cycle scanning and advertise only when necessary.
Movement	A critical relay walks away, creating a partition.	Anchor the backbone to powered, mapped positions.
Platform fragmentation	Different devices expose different P2P features.	Capability negotiation and transport fallback.
Crowd density	Many phones advertising causes collision and discovery delay.	Random jitter, sparse updates and small crew scopes.
Privacy concerns	Continuous discovery and location sharing reduces trust.	Explicit crew consent, rotating identifiers and visible controls.

Acceptable phone relay role

Loc8 may implement store-and-forward for a crew as an optional feature: a foreground phone receives an encrypted message with a short expiry and forwards it to an intended crew member. The user interface should label delivery as pending until acknowledged. Do not silently represent an opportunistic hop as guaranteed delivery.

Platform hierarchy

Priority	Transport	Use
1	Wi-Fi Aware / equivalent secure P2P	Higher-rate direct exchange where both devices support it.
2	Google Nearby Connections or approved cross-platform abstraction	Offline discovery and encrypted exchange across available radios.
3	Venue Wi-Fi / local LAN	High capacity when the venue supplies a suitable network.
4	BLE GATT direct link	Small groups, configuration and fallback.
5	BLE advertising	Discovery, tiny status and delayed opportunistic messages.

FROM RADIO CELLS TO VENUE ZONES

9. Arena deployment design

Initial node counts

Venue type	Starting count	Placement concept	Validation
Small hall / warehouse	4-6	Corners plus centre or entrances	Walk test with crowd simulation
Indoor arena	6-8	Bowl perimeter, floor edge, concourses and gateway	Test empty and occupied
Large stadium	8-16+	Multiple stand sectors, floor, concourse rings, entrances	Professional RF survey
Festival site	Per-zone design	Stages, bars, welfare, gates and routes	Weather and temporary rigging review

Placement rules

- Elevate antennas above typical body height and preserve line of sight where possible.
- Use plastic or RF-transparent enclosures. Do not place the antenna inside a metal box.
- Keep nodes away from large video walls, metal truss clusters and high-power 2.4 GHz access points where practical.
- Create 20-30% overlap between measured usable cells so one failure does not create an immediate hole.
- Give concourse bends, stairs, gates, backstage and welfare areas dedicated coverage.
- Use tamper-resistant professional mounts and safety bonds. Venue rigging must be performed by authorised personnel.

Gateway and zone model

Assign stable zone identifiers such as FLOOR-A, BOWL-NORTH, CONCOURSE-EAST and WELFARE-1. Each relay publishes health, firmware version and last-seen neighbour data. The gateway maps those identifiers to the Loc8 venue map and can issue signed, expiring messages to one zone, a group of zones or all zones.

COVERAGE ACCEPTANCE CRITERION

Do not accept a deployment because one phone received one packet at the boundary. Define a repeatable target, such as at least 95% successful zone detections within 5 seconds across the mapped public area, measured on a representative iOS and Android device set under realistic occupancy.

PROTOTYPE KIT AND VENUE PILOTS

10. Cost model

Prices below are research snapshots or planning allowances dated 21 July 2026. VAT treatment is shown where the source permits. Shipping, labour, testing, venue access and certification are excluded unless stated.

One DIY long-range relay node

Item	Basis	Budget inc VAT
Nordic nRF21540 Development Bundle	Farnell listing £63.64 ex VAT	£76.37
USB cable	Not supplied with DK	£4-£7
Plastic IP-rated enclosure	Allowance; size to be verified	£10-£15
Standoffs, glands, vent and fixings	Allowance	£8-£15
5 V USB supply	Bench / indoor prototype	£8-£12
or PoE splitter	RobotShop example £22.42 inc VAT	£20-£30
Mount and safety bond	Temporary prototype allowance	£10-£25
Indicative total	USB or PoE choice changes total	£105-£145

System budget ranges

Kit	Hardware concept	Planning budget
Four-node lab / small hall	4 relays, one basic gateway, enclosures, power and spares	£650-£800
Six-node arena pilot	6 relays, gateway, PoE/power, mounts and spares	£900-£1,200
Eight-node arena	8 relays, gateway, mounts, power and spares	£1,200-£1,600
Twelve-node stadium pilot	12 relays, gateway, additional mounts/cabling and spares	£1,800-£2,500
Sixteen-node stadium	16 relays, gateway, mounts, power and spares	£2,400-£3,200

These totals assume Loc8 engineering time is not costed. They are suitable for experiment planning, not a commercial quotation.

WHAT THE HARDWARE TOTAL LEAVES OUT

11. Commercialisation costs beyond the prototype

Cost area	Why it exists	Action before launch
RF and EMC pre-compliance	A board that works on a bench may fail emissions, immunity or antenna performance tests.	Book a pre-scan with an accredited laboratory.
Radio Equipment Regulations	GB radio products must meet safety, EMC and efficient spectrum-use essential requirements.	Create technical documentation and a Declaration of Conformity route.
Bluetooth qualification	Products using Bluetooth technology must follow the Bluetooth SIG qualification process.	Choose qualified components and obtain a current fee/qualification plan.
Custom PCB engineering	A production node should not contain a whole development bundle.	RF layout, antenna match, DFM and factory test design.
Enclosure and tooling	Venue products need flame, ingress, impact, thermal and mounting decisions.	Industrial design and certified materials review.
Cybersecurity and privacy	Location data and radio control create abuse and data protection risks.	Threat model, penetration test, DPIA and retention policy.
Venue deployment labour	Rigging, cabling, surveys and show-day support are material costs.	Price per venue and per event separately from hardware.
Insurance and support	Public venue use creates operational liability.	Product liability, maintenance and incident process.

Production unit economics

A custom PCB can eventually cost less than a development bundle, but any early '£35-£65 BOM' estimate would be highly sensitive to radio module choice, antenna, enclosure, PoE, volumes and test requirements. Loc8 should request contract manufacturer quotations only after the prototype RF survey fixes the required link budget and enclosure format.

BUDGET GATE

Keep certification, tooling, production software, mobile app development and professional venue installation outside the £105-£145 node figure. They are separate programmes and may exceed the prototype hardware cost by orders of magnitude.

NORDIC NRF21540 DEVELOPMENT BUNDLE PROTOTYPE

12. DIY relay build guide

PROTOTYPE SCOPE

This build is suitable for controlled laboratory and venue trials. It is not a certified product, weatherproof commercial installation or authorised rigging design.

Parts list

Part	Specification / note
nRF21540 Development Bundle	Use the full nRF21540 DK board with nRF52840 and front-end module.
Micro-USB data cable	Nordic says it is not supplied. Use a known data-capable cable.
5 V supply or PoE splitter	Use a reputable protected supply. Confirm current and connector polarity.
Plastic enclosure	Leave clearance around the selected onboard antenna.
Nylon standoffs	Avoid board contact with conductive surfaces.
Cable gland / strain relief	Match the power cable and required ingress protection.
Mounting plate and safety bond	Must comply with venue rigging practice.
Laptop and test phones	Use multiple iOS and Android models.

Assembly

1. Inspect and bench-test
 - Power the unmodified DK on a non-conductive bench. Confirm the debugger enumerates and run the vendor's basic sample before enclosure work.
2. Choose the antenna path
 - Use the board documentation to select the intended onboard antenna. Do not attach an external antenna unless the RF path, gain and resulting EIRP have been reviewed.
3. Prepare the enclosure
 - Mark standoff and gland positions. Keep the antenna end away from batteries, cable bundles, metal fixings and the mounting plate.
4. Mount without stress
 - Use nylon standoffs and washers. Ensure buttons, USB and debug access remain serviceable.
5. Add power
 - For bench trials use USB. For a venue pilot, PoE can simplify cable runs, but the splitter and enclosure thermal behaviour must be validated.
6. Label the node
 - Add node ID, zone, MAC/device identifier, firmware version, power requirement and an engineering contact.
7. Close and retest
 - Measure receive performance before and after closing the enclosure. A large change indicates antenna detuning or shielding.

BEACON FIRST, MESH SECOND

13. Firmware build guide

Development environment

Install Nordic's current nRF Connect SDK through the vendor-supported toolchain and Visual Studio Code integration. Start with a known sample before adding Loc8 protocol code. The nRF21540 DK board target is documented by Zephyr as nrf21540dk/nrf52840.

Stage A: basic iBeacon-style transmitter

Use the Zephyr iBeacon sample to validate advertising and receiver range. Build from the nRF Connect SDK terminal:

```
west build -b nrf21540dk/nrf52840 zephyr/samples/bluetooth/ibeacon
```

Flash with the supported VS Code action or the standard command below after confirming the correct connected board:

```
west flash
```

Loc8 identifier plan

Field	Prototype use
128-bit UUID	Loc8 organisation / protocol identifier
Major	Venue identifier
Minor	Zone or node identifier
Measured power	Calibrated RSSI at the selected reference distance
Rotating token	Use a service-data frame in the later signed Loc8 protocol; do not expose stable user identity

Front-end module gain

Nordic documents the build option CONFIG_MPSL_FEM_NRF21540_TX_GAIN_DB. The documented default is 10 dB, while the device has factory-calibrated output modes associated with 20 dBm and 10 dBm. Begin at the default, measure conducted/radiated output with appropriate equipment, then increase only after confirming the complete system EIRP and compliance constraints.

```
CONFIG_MPSL_FEM_NRF21540_TX_GAIN_DB=10
```

FROM A BEACON TO A MANAGED VENUE NETWORK

14. Mesh firmware and gateway design

Provisioning

Each fixed node needs a unique device identity and secure provisioning path. Commission nodes before the event or through an authenticated local process. Store separate network, application and device keys. Do not hard-code one global production key into firmware images.

Minimum node models

Model / function	Purpose	Suggested data
Health Server	Report faults and attention state	Self-test, resets and fault code
Generic OnOff or vendor control	Enable maintenance modes	Radio test, identify LED and reboot request
Vendor telemetry	Publish operational data	Firmware, temperature, power and neighbours
Zone broadcast control	Set local advertisement payload	Zone ID, event epoch and signed message
DFU path	Authorised firmware update	Signed image and rollback protection

Gateway responsibilities

- Maintain the venue zone map and node inventory.
- Show last-seen time, RSSI/quality trends, power state and firmware version.
- Sign and expire control messages; reject replayed or unauthorised commands.
- Rate-limit publications to protect mesh airtime.
- Continue locally if the internet connection fails.
- Export an audit log without storing attendee position history by default.

Suggested Loc8 broadcast envelope

```
version | event_id | zone_id | epoch | message_type | short_payload | signature_tag
```

The exact packing, cryptography and key rotation require a security review. Keep advertisements anonymous and non-trackable. Private crew messages should be end-to-end encrypted and addressed with rotating identifiers.

EVIDENCE BEFORE PROMISES

15. Range, load and venue test plan

Phase 1: bench

1. Firmware repeatability
Flash three nodes and confirm identical configuration, unique IDs and recoverable updates.
2. Power behaviour
Test USB and PoE power-up, brownout, reconnect and 24-hour operation.
3. Thermal check
Measure enclosure temperature in expected ambient conditions.
4. Security basics
Confirm unauthorised configuration and replayed messages are rejected.

Phase 2: open-space range

Place the node at the intended deployment height. Test at 25, 50, 75, 100 and 125 m on multiple phones. At each point record packet detection latency, packet count over 60 seconds, RSSI distribution, phone orientation and whether the phone is foregrounded, locked or backgrounded.

Phase 3: obstructed venue

Repeat measurements behind seating, around concourse bends, at floor level, inside pockets and with people between transmitter and receiver. Test while the venue Wi-Fi and normal production radios are operating.

Phase 4: traffic and failure

Test	Method	Pass concept
Mesh load	Increase publications and relays in steps	Latency and delivery remain inside target
Node loss	Power off the most connected relay	Coverage degrades gracefully or reroutes
Gateway loss	Disconnect gateway and internet separately	Local broadcasts continue safely
Crowd simulation	Use people or RF-absorbing placement	Mapped detection criterion remains met
Phone matrix	Representative iOS and Android devices	Documented support and fallback behaviour
Recovery	Restore node and rotate configuration	Automatic authenticated return to service

RECORD RESULTS CONSISTENTLY

16. Measurement sheet and acceptance criteria

Field	Record
Event / venue / date	Name, location and UTC/local time
Node	ID, firmware, antenna, height, power and FEM gain
Receiver	Phone model, OS, app version and battery state
Position	Zone, coordinates/map point, range and orientation
Environment	Occupancy, doors, stage, Wi-Fi load and weather if outdoors
Mode	Foreground, background, locked, screen-on or active P2P
Outcome	Time to first detection, packets/minute, RSSI median and loss
Notes	Obstruction, unexpected interference and photographs

Proposed pilot gates

Gate	Example criterion
Coverage	At least 95% of mapped public points detect the correct zone within 5 seconds.
Overlap	Every critical public route sees at least two viable nodes.
Resilience	Loss of any one relay does not isolate a welfare, exit or principal public zone.
Battery / power	No unplanned resets in a 12-hour event profile plus setup margin.
Phone impact	Background power draw stays within the agreed app budget.
Privacy	No stable attendee identifier is observable in plaintext over the air.
Operations	A trained operator can identify, isolate and replace a failed node.

The numbers above are proposed engineering gates. Loc8 should revise them after the first venue pilot and define separate targets for zone detection, friend ranging and message delivery.

PROTOTYPE FREEDOM IS NOT PRODUCT APPROVAL

17. UK spectrum and product compliance

2.4 GHz spectrum

Ofcom Interface Requirement IR 2030 lists wideband data transmission systems in 2400-2483.5 MHz at up to 100 mW EIRP, subject to mitigation techniques and the applicable standard, including EN 300 328. Bluetooth equipment must be configured so the complete transmitter, antenna and duty behaviour remains compliant. Do not equate chip output power with permitted system EIRP.

Placing hardware on the GB market

The Radio Equipment Regulations 2017 require radio equipment placed on the Great Britain market to meet essential requirements covering safety, electromagnetic compatibility and efficient use of spectrum. Current UK guidance explains the manufacturer's technical documentation, conformity assessment, declaration, marking, traceability and corrective-action responsibilities. CE marking continues to be recognised for relevant products in Great Britain under current policy, but Loc8 must obtain product-specific compliance advice.

Compliance workstream

Stage	Output
Classification	Intended use, markets, radio bands and applicable legislation/standards
Design controls	Qualified module choice, antenna specification, power limits and safety design
Pre-compliance	RF, EMC, electrical safety and environmental engineering tests
Formal assessment	Laboratory reports and any required approved/notified body route
Bluetooth SIG	Membership, qualification and declaration process for the branded product
Technical file	Schematics, BOM, risk assessment, test reports, instructions and labels
Production control	Factory test, change control, traceability and field corrective action

NO HIGH-POWER SHORTCUT

An external RF amplifier or high-gain antenna can change the EIRP, spurious emissions and conformity basis. Do not add one to a certified module or development board without qualified RF and compliance review.

LOCATION PRODUCTS NEED TRUST BY DESIGN

18. Security, privacy and safeguarding

Threat	Example	Required control
Tracking	Static BLE identifier follows a person between venues.	Rotating pseudonyms and no user identity in advertisements.
Spoofing	Attacker pretends to be a friend or a venue node.	Mutual authentication and signed infrastructure messages.
Replay	Old lost-mode alert is retransmitted.	Epoch, nonce, short expiry and replay cache.
Eavesdropping	Nearby listener observes crew positions.	End-to-end encryption with crew-scoped keys.
Denial of service	Flooded advertisements or mesh traffic block legitimate data.	Rate limits, traffic priorities and anomaly monitoring.
Compromised node	Stolen relay leaks network credentials.	Secure storage, per-device keys, revocation and signed firmware.
Coercive use	An abusive person pressures someone to share location.	Per-person consent, pause/leave controls and discreet safety UX.

Data minimisation

The fixed venue layer does not need attendee GPS history. It should know infrastructure health and zone configuration. Crew location data can remain on participating phones where possible, with clear freshness indicators and automatic deletion after the event. If Loc8 later adds cloud synchronisation, make it a separate, consented service with defined retention.

Safety boundary

Loc8 can help friends find each other, but Bluetooth availability is not guaranteed. The app must not replace emergency services, steward instructions, public address systems, regulated assistive-listening systems or venue incident procedures. Lost mode should always show official venue help points and emergency guidance supplied by the operator.

EVIDENCE-LED STEPS

19. Recommended Loc8 delivery roadmap

Phase	Duration concept	Deliverable	Decision gate
0. Requirements	1-2 weeks	Use cases, privacy model, target venues and success metrics	Approve system boundary
1. Bench prototype	2-4 weeks	3 nRF21540 nodes, beacon firmware and measurement app	Stable repeatable range data
2. Small hall pilot	2-4 weeks	4-6 fixed nodes, gateway dashboard and zone detection	Coverage and background feasibility
3. Hybrid P2P prototype	4-8 weeks	iOS/Android capability negotiation and crew exchange	Cross-platform delivery and battery targets
4. Arena pilot	Event dependent	6-8 nodes in a real venue with operational monitoring	Acceptance criteria met under occupancy
5. Production design	3-6+ months	Custom hardware, enclosure, manufacturing and compliance	Certified pilot batch
6. Venue product	Commercial	Support, installation method, training and SLAs	Repeatable deployment economics

First purchase recommendation

Buy three nRF21540 Development Bundles, suitable plastic enclosures, data cables and one PoE splitter for lab work. Three nodes are enough to test a two-hop path and observe the effect of relaying. Then expand to six nodes only after the range and enclosure results are recorded.

Questions the first pilot must answer

- What is the 95th-percentile time to zone detection on supported phones?
- How much does a dense crowd reduce the measured usable radius?
- Does Wi-Fi Aware / Nearby provide acceptable cross-platform crew exchange?
- What update interval gives useful friend radar without excessive battery use?
- How many nodes and which locations remove critical dead zones?
- What is the install and support cost per venue day?

GO / NO-GO RULE

Do not invest in a custom PCB until the six-node pilot proves the product experience and defines the necessary RF performance. Hardware miniaturisation before that point risks optimising the wrong architecture.

PLAIN-LANGUAGE COMPANY RESPONSE

20. Ready-to-use answer: 'OK, and now that covers large areas?'

“ Yes. Loc8 covers a large area by installing several powered Bluetooth relay nodes around the venue. Each node covers its own local zone, and the nodes pass small messages between zones, so the network reaches across the arena in hops. Phones do not all connect to one box, and the system does not depend on every attendee acting as a relay. A typical indoor arena would begin with about six to eight fixed nodes, while a stadium may need eight to sixteen or more after an RF survey. The exact count comes from measured coverage, not an advertised maximum range.

Fast factual answers

Question	Answer
How big is 100 m?	About one football-pitch length. A 100 m radius creates a 200 m diameter circle covering about 31,416 m ² or 7.76 acres.
How many connect to one Bluetooth device?	For direct maintained links, about 20 is an absolute nRF52840 stack limit and a robust product would normally use fewer. Broadcast listeners do not connect, so thousands can listen.
What does mesh do?	Relay nodes repeat messages over hops. It extends the network footprint; it does not amplify one transmission.
How many people on mesh?	32,767 is the formal element address ceiling, but not a safe attendee-capacity claim. Keep phones as receivers/endpoints and fixed boxes as relays.
Will phones relay?	They can opportunistically, especially in the foreground, but mobile OS background controls make them unsuitable as the guaranteed backbone.
What does one node cost?	About £105-£145 for the recommended DIY prototype node, excluding Loc8 engineering, compliance and venue labour.

WHAT MANAGEMENT SHOULD TRACK

21. Principal risks and decisions

Risk	Likelihood	Impact	Current response
Crowd reduces range below plan	High	High	Conservative cells, elevation, overlap and occupied tests
iOS/Android background inconsistency	High	High	Fixed infrastructure and capability-based fallbacks
Mesh congestion from overuse	Medium	High	Sparse control data, zones, TTL and wired backhaul
Privacy perception blocks adoption	Medium	High	Rotating IDs, explicit consent and local-first data
Prototype mistaken for compliant product	Medium	High	Clear phase gate and compliance workstream
Venue installation cost erodes margin	Medium	Medium	PoE, repeatable mounting kit and per-venue pricing
Hardware built before UX is proven	Medium	High	Three-node and six-node pilots before custom PCB
Overclaim creates contractual exposure	Medium	High	Publish measured conditions and avoid guaranteed range

Decisions required from Loc8

- Choose whether venue infrastructure is core to the business or a premium add-on.
- Define the supported phone and operating-system baseline, including iOS 26 Wi-Fi Aware availability.
- Set privacy promises before defining the radio payload.
- Select the first representative pilot venue and secure installation permission.
- Appoint owners for mobile networking, embedded firmware, RF/compliance and venue operations.
- Approve a hardware-only six-node pilot allowance of £900-£1,200 plus engineering and venue costs.

TERMS USED IN THIS REPORT

Appendix A. Glossary

Term	Meaning
Advertisement	A short BLE transmission that can be received without forming a connection.
Auracast	Bluetooth broadcast audio capability. Useful venue precedent, but not a general data mesh.
Backhaul	The path connecting infrastructure nodes to a gateway, often Ethernet or another network.
BLE	Bluetooth Low Energy.
Coded PHY	BLE physical-layer modes that trade data rate for improved sensitivity and range.
EIRP	Effective isotropic radiated power, including transmitter power, antenna gain and losses.
Element	An independently addressable entity within a Bluetooth Mesh node.
FEM	Front-end module that can add transmit gain and receive gain around the radio.
GATT	Bluetooth data model used for services and characteristics over a connection.
Gateway	Local computer or node that controls and observes the venue infrastructure.
Managed flooding	Mesh method in which eligible relays retransmit messages under configured controls.
Node	A device that participates in a Bluetooth Mesh network; it may contain one or more elements.
PoE	Power over Ethernet.
Proxy	A mesh feature that lets a non-mesh BLE device communicate with a mesh through a proxy connection.
RSSI	Received Signal Strength Indicator. Useful but noisy and not a direct distance measurement.
TTL	Time To Live, reduced at each mesh hop to limit propagation.
Wi-Fi Aware	Direct nearby discovery and peer-to-peer communication without an access point or internet.

INTERPRET FIGURES CORRECTLY

Appendix B. Assumptions and calculation notes

Geometry

For a nominal 100 m radius: diameter = $2 \times 100 = 200$ m. Circular area = $\pi \times \text{radius squared} = 3.1416 \times 10,000 = 31,416$ m². Acres = $31,416 / 4,046.86 = 7.76$ acres.

Prices

The nRF21540 Development Bundle price uses £63.64 ex VAT from the cited Farnell listing. At 20% VAT this equals £76.37. Accessory figures are allowances. Vendor pages and distributors may change prices, stock and tax treatment after the research date.

Node counts

Six to eight nodes for an indoor arena and eight to sixteen or more for a stadium are early planning ranges derived from a cell-and-overlap architecture. They are not produced by a detailed floor plan or propagation model. Concourses, stages, floor density and installation height can materially increase the count.

Capacity

'80,000+' describes the architectural fact that passive broadcast reception is not limited by maintained BLE connection slots. It does not assert that every one of 80,000 phones receives every packet. Detection probability is affected by RF conditions, receiver diversity, scan scheduling and application state.

Connection limit

The cited 20-link figure is an absolute supported limit discussed for Nordic nRF52840 production controller stacks. Controller, role, connection parameters, RAM, throughput and application demands determine the usable number. Zephyr includes demonstration samples with higher central-role link counts, but these are not a reason to design an arena service around dozens of direct phone connections.

Technology freshness

Wi-Fi Aware on Apple platforms is current as of the research date. Apple's technical note states iOS 26 and iPhone 12 or later. Adoption and API behaviour should be rechecked against the release and App Store submission target.

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MOBILE PLATFORMS, SPECTRUM AND COMPLIANCE

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SOURCE USE

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